GRADE 7 ACCELERATED · UNIT 6 · LESSONS 1–2

Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Independent Set 6.01 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. Can sides of 4 cm, 8 cm and 13 cm make a triangle?

No — 4 + 8 = 12 is less than 13.

2. Can sides of 9 inches, 9 inches and 18 inches make a triangle?

No — 9 + 9 = 18 lies flat.

The equal case is not a triangle.

3. Two sides of a triangle are 5 m and 11 m. Write an inequality for the length x, in meters, of the third side.

6 < x < 16

11 − 5 and 11 + 5; never equal.

4. Do sides of 28 cm, 45 cm and 53 cm make a right triangle?

Yes — 784 + 2,025 = 2,809 = 53^2.

5. Do sides of 17 ft, 10 ft and 14 ft make a right triangle?

No — $10^2 + 14^2 = 296$, but $17^2 = 289$.

The longest side is listed first.

6. A carpenter joins boards of 30 inches, 40 inches and 50 inches into a triangle. Is one corner a right angle?

Yes — 900 + 1,600 = 2,500 = 50^2.

Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Question Bank 6.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, can the three lengths make a triangle?

1. 3 m, 9 m, 11 m

Yes — 3 + 9 = 12 > 11

2. 5 inches, 13 inches, 7 inches

No — 5 + 7 = 12 < 13

The longest is listed second.

3. 8 ft, 4 ft, 4 ft

No — 4 + 4 = 8 lies flat

The equal case.

For questions 4 and 5, two sides of a triangle are given. Write an inequality for the third side, x.

4. 3 cm and 8 cm

5 < x < 11 (centimeters)

5. 1 cm and 5 cm; the third side is a whole number of centimeters — which one?

4 < x < 6, so x = 5 cm — the only whole number between

For questions 6 through 8, do the lengths make a right triangle?

6. 4.5 cm, 6 cm, 7.5 cm

Yes — $20.25 + 36 = 56.25 = 7.5^2$

7. 26 inches, 24 inches, 10 inches

Yes — $10^2 + 24^2 = 676 = 26^2$

The longest is listed first.

8. 7 m, 10 m, 12 m

No — 49 + 100 = 149, not 144

Choose one.

9. Which set of lengths makes a right triangle?

- ○ A. 4, 5, 6
- ○ B. 2, 3, 5
- ○ C. 6, 8, 12
- ● **D. 11, 60, 61**

D

121 + 3,600 = 3,721 = 61^2.

Triangle Inequality Theorem; Converse of the Pythagorean Theorem

Question Bank - Additional 6.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, can the three lengths make a triangle?

1. 4 m, 10 m, 12 m

Yes — 4 + 10 = 14 > 12

2. 6 inches, 17 inches, 9 inches

No — 6 + 9 = 15 < 17

The longest is listed second.

3. 14 ft, 7 ft, 7 ft

No — 7 + 7 = 14 lies flat

The equal case.

For questions 4 and 5, two sides of a triangle are given. Write an inequality for the third side, x.

4. 6 cm and 9 cm

3 < x < 15 (centimeters)

5. 1 inch and 8 inches; the third side is a whole number of inches — which one?

7 < x < 9, so x = 8 in — the only whole number between

For questions 6 through 8, do the lengths make a right triangle?

6. 2.4 m, 3.2 m, 4 m

Yes — 5.76 + 10.24 = 16 = 4^2

7. 39 inches, 36 inches, 15 inches

Yes — $15^2 + 36^2 = 1{,}521 = 39^2$

The longest is listed first.

8. 8 m, 9 m, 12 m

No — 64 + 81 = 145, not 144

Choose one.

9. Which set of lengths makes a right triangle?

- ○ A. 5, 6, 7
- ● **B. 14, 48, 50**
- ○ C. 9, 12, 16
- ○ D. 3, 4, 7

B

196 + 2,304 = 2,500 = 50^2.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

Pythagorean Theorem

Independent Set 6.03 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A right triangle has legs of 12 m and 5 m. How long is the hypotenuse?

13 m

144 + 25 = 169 = 13^2.

2. A right triangle has legs of 3 ft and 8 ft. How long is the hypotenuse, exactly and to the nearest tenth?

√73 ft ≈ 8.5 ft

9 + 64 = 73.

3. A right triangle has a hypotenuse of 39 cm and one leg of 36 cm. How long is the other leg?

15 cm

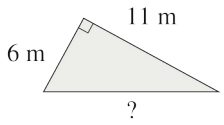
1,521 − 1,296 = 225 = 15^2.

4. A right triangle has a hypotenuse of 11 inches and one leg of 4 inches. How long is the other leg, exactly and to the nearest tenth?

√105 in ≈ 10.2 in

121 − 16 = 105.

5. The right triangle on your paper has legs of 6 m and 11 m. How long is its hypotenuse, to the nearest tenth?



√157 m ≈ 12.5 m

36 + 121 = 157; c is across from the square corner.

6. A right triangle has a hypotenuse of 25 yd and one leg of 24 yd. The other leg is x. Write the equation that finds x, then find x.

$24^2 + x^2 = 25^2$; x = 7 yd

625 − 576 = 49 = 7^2.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

Pythagorean Theorem

Question Bank 6.03 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, find the hypotenuse. Give it exactly, and to the nearest tenth when it is not a whole number.

1. Legs of 8 cm and 6 cm

10 cm

2. Legs of 4 inches and 11 inches

√137 in ≈ 11.7 in

16 + 121 = 137.

3. Legs of 7 m and 8 m

√113 m ≈ 10.6 m

49 + 64 = 113.

For questions 4 through 6, find the missing leg. Give it exactly, and to the nearest tenth when it is not a whole number.

4. Hypotenuse 13 ft, one leg 12 ft

5 ft

169 − 144 = 25.

5. Hypotenuse 8 cm, one leg 5 cm

√39 cm ≈ 6.2 cm

64 − 25 = 39.

6. Hypotenuse 61 inches, one leg 60 inches

11 in

3,721 − 3,600 = 121 = 11^2.

7. A rectangle is 5 inches wide and 12 inches long. How long is its diagonal?

13 in

The diagonal is the hypotenuse of a 5-by-12 right triangle.

Choose one.

8. A right triangle has a hypotenuse of 25 m and one leg of 24 m. How long is the other leg?

- ○ A. 1 m
- ○ B. 34.7 m
- ● **C. 7 m**
- ○ D. 49 m

C

625 − 576 = 49 = 7^2.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

Pythagorean Theorem

Question Bank - Additional 6.03 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 3, find the hypotenuse. Give it exactly, and to the nearest tenth when it is not a whole number.

1. Legs of 9 ft and 12 ft

15 ft

2. Legs of 3 inches and 10 inches

√109 in ≈ 10.4 in

9 + 100 = 109.

3. Legs of 6 m and 13 m

√205 m ≈ 14.3 m

36 + 169 = 205.

For questions 4 through 6, find the missing leg. Give it exactly, and to the nearest tenth when it is not a whole number.

4. Hypotenuse 10 cm, one leg 8 cm

6 cm

100 − 64 = 36.

5. Hypotenuse 7 inches, one leg 4 inches

√33 in ≈ 5.7 in

49 − 16 = 33.

6. Hypotenuse 37 m, one leg 35 m

12 m

1,369 − 1,225 = 144 = 12^2.

7. A rectangle is 6 cm wide and 8 cm long. How long is its diagonal?

10 cm

The diagonal is the hypotenuse of a 6-by-8 right triangle.

Choose one.

8. A right triangle has a hypotenuse of 26 ft and one leg of 24 ft. How long is the other leg?

- ○ A. 100 ft
- ● **B. 10 ft**
- ○ C. 2 ft
- ○ D. 35.4 ft

B

676 − 576 = 100 = 10^2.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

Pythagorean Theorem in Real-World Problems

Independent Set 6.04 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A tablet screen is 5 inches wide and 8 inches tall. How long is its diagonal, to the nearest tenth of an inch?

About 9.4 in

25 + 64 = 89; √89 ≈ 9.43.

2. A fire truck's 50-ft ladder reaches 48 ft up a building. How far from the building is its foot?

14 ft

2,500 − 2,304 = 196; the ladder is c.

3. A ramp rises 3 ft over a run of 14 ft. How long is the ramp, to the nearest tenth of a foot?

About 14.3 ft

9 + 196 = 205; √205 ≈ 14.32.

4. A park is 7 m by 11 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides? Give it to the nearest tenth of a meter.

About 5.0 m shorter

Along two sides: 18 m. Across: √170 ≈ 13.04 m. 18 − 13.04 = 4.96, so 5.0.

5. Will a 10-in pencil lie flat in a box 7 inches by 8 inches, corner to corner?

Yes — the diagonal is about 10.6 in.

49 + 64 = 113; √113 ≈ 10.63 > 10.

6. A kite flies on 19 m of tight string. Straight below the kite, level with the hand that holds the string, is a point 12 m from the hand. How high is the kite above that point, to the nearest tenth of a meter?

About 14.7 m

361 − 144 = 217; √217 ≈ 14.73; the string is c.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

Pythagorean Theorem in Real-World Problems

Question Bank 6.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 7, sketch the right triangle first. Round to the nearest tenth when the answer is not a whole number.

1. A TV screen is 9 inches tall and 12 inches wide. How long is its diagonal?

15 in

81 + 144 = 225.

2. A 14-ft ladder leans on a wall with its foot 3 ft from the wall. How high does it reach?

About 13.7 ft

196 − 9 = 187; √187 ≈ 13.67.

3. A ramp rises 2 m over a run of 9 m. How long is the ramp?

About 9.2 m

4 + 81 = 85; √85 ≈ 9.22.

4. A courtyard is 5 m by 13 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides?

About 4.1 m

Along two sides 18 m; across √194 ≈ 13.93 m; 18 − 13.93 = 4.07.

5. Will a 12-in ruler lie flat in a box 6 inches by 11 inches, corner to corner?

Yes — the diagonal is about 12.5 in.

36 + 121 = 157; √157 ≈ 12.53, so 12.5.

6. A kite's 13-m string is held at a spot 5 m from the point straight below the kite. How high is the kite?

12 m

169 − 25 = 144; the string is c.

7. A wire runs from the top of an 8-m pole to a stake in the ground 6 m from the pole's base. How long is the wire?

10 m

64 + 36 = 100.

Choose one.

8. A 13-ft ladder reaches 12 ft up a wall. How far from the wall is its foot?

- ○ A. 1 ft
- ● **B. 5 ft**
- ○ C. 17.7 ft
- ○ D. 25 ft

B

$169 - 144 = 25 = 5^2$.

9. A ramp rises 3 ft over a run of 20 ft. Which length is the hypotenuse?

- ○ A. the run
- ○ B. the rise
- ● **C. the ramp**
- ○ D. none of them

C

The square corner is under the high end; across from it is the ramp.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

Pythagorean Theorem in Real-World Problems

Question Bank - Additional 6.04 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 7, sketch the right triangle first. Round to the nearest tenth when the answer is not a whole number.

1. A window is 8 ft wide and 6 ft tall. How long is its diagonal?

10 ft

64 + 36 = 100.

2. An 11-ft ladder leans on a wall with its foot 4 ft from the wall. How high does it reach?

About 10.2 ft

121 − 16 = 105; √105 = 10.246…, so 10.2.

3. A ramp rises 3 ft over a run of 10 ft. How long is the ramp?

About 10.4 ft

9 + 100 = 109; √109 ≈ 10.44.

4. A garden bed is 3 m by 4 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides?

2 m

Along two sides 7 m, across 5 m.

5. Will a 15-in rod lie flat in a box 9 inches by 11 inches, corner to corner?

No — the diagonal is only about 14.2 in.

81 + 121 = 202; √202 ≈ 14.21.

6. A kite's 17-m string is held at a spot 8 m from the point straight below the kite. How high is the kite?

15 m

289 − 64 = 225; the string is c.

7. A wire runs from the top of a 9-m pole to a stake in the ground 12 m from the pole's base. How long is the wire?

15 m

81 + 144 = 225.

Choose one.

8. A 10-ft ladder reaches 8 ft up a wall. How far from the wall is its foot?

- ● **A. 6 ft**
- ○ B. 2 ft
- ○ C. 36 ft
- ○ D. 12.8 ft

A

$100 - 64 = 36 = 6^2$.

9. A wire runs from the top of a pole to a stake in the ground. Which length is the hypotenuse?

- ○ A. the pole
- ● **B. the wire**
- ○ C. the ground from the pole to the stake
- ○ D. none of them

B

The pole stands square on the ground; across from that corner is the wire.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 5

Distance on the Coordinate Plane

Independent Set 6.05 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. Find the distance between $(-6, 3)$ and $(7, 3)$.

13 units

Same y: 7 − (−6) = 13.

2. Find the distance between $(5, -7)$ and $(5, 4)$.

11 units

Same x: 4 − (−7) = 11.

3. Find the distance between $(-7, 1)$ and $(5, -4)$.

13 units

Across 12, down 5; 144 + 25 = 169.

4. Find the distance between $(-2, 5)$ and $(1, -2)$, to the nearest tenth.

$\sqrt{58} \approx 7.6$ units

Across 3, down 7; 9 + 49 = 58.

5. How far is $(-4, 9)$ from the origin, to the nearest tenth?

$\sqrt{97} \approx 9.8$ units

Legs 4 and 9; 16 + 81 = 97.

6. On a map in kilometers, a pool is at $(-2, -3)$ and a park at $(8, 8)$. How far apart are they in a straight line, to the nearest tenth of a kilometer?

$\sqrt{221} \approx 14.9$ km

Across 10, up 11; 100 + 121 = 221.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 5

Distance on the Coordinate Plane

Question Bank 6.05 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 7, find the distance between the two points. Round to the nearest tenth when it is not a whole number.

1. $(-8, 4)$ and $(2, 4)$

10 units

Same y.

2. $(3, -6)$ and $(3, 5)$

11 units

Same x.

3. $(-1, -1)$ and $(2, 3)$

5 units

Across 3, up 4.

4. $(-6, 2)$ and $(6, -3)$

13 units

Across 12, down 5.

5. $(-3, -2)$ and $(4, 1)$

$\sqrt{58} \approx 7.6$ units

Across 7, up 3.

6. $(0, 0)$ and $(-5, 7)$

$\sqrt{74} \approx 8.6$ units

Legs 5 and 7.

7. $(2, -4)$ and $(-1, 6)$

$\sqrt{109} \approx 10.4$ units

Across 3, up 10.

Choose one.

8. What is the distance between $(-6, -1)$ and $(2, -1)$?

- ○ A. 4
- ○ B. −8
- ○ C. 9
- ● **D. 8**

D

Same y: 2 − (−6) = 8.

9. Which gives the distance between $(5, 2)$ and $(-2, -3)$?

- ● **A.** $\sqrt{7^2 + 5^2}$
- ○ B. $\sqrt{3^2 + 5^2}$
- ○ C. $\sqrt{5^2 + 2^2}$
- ○ D. $7 + 5$

A

Across 7, down 5: √74.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 5

Distance on the Coordinate Plane

Question Bank - Additional 6.05 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 7, find the distance between the two points. Round to the nearest tenth when it is not a whole number.

1. $(-5, -2)$ and $(4, -2)$

9 units

Same y.

2. $(-1, 7)$ and $(-1, -6)$

13 units

Same x.

3. $(-4, 1)$ and $(2, 9)$

10 units

Across 6, up 8.

4. $(-3, -5)$ and $(6, 7)$

15 units

Across 9, up 12.

5. $(-2, 3)$ and $(4, -2)$

$\sqrt{61} \approx 7.8$ units

Across 6, down 5.

6. $(0, 0)$ and $(8, -3)$

$\sqrt{73} \approx 8.5$ units

Legs 8 and 3.

7. $(-5, 4)$ and $(2, -5)$

$\sqrt{130} \approx 11.4$ units

Across 7, down 9.

Choose one.

8. What is the distance between $(-7, 3)$ and $(-7, -4)$?

- ○ A. 8
- ○ B. −7
- ○ C. 1
- ● **D. 7**

D

Same x: 3 − (−4) = 7.

9. Which gives the distance between $(-4, 5)$ and $(0, -4)$?

- ○ A. $\sqrt{4^2 + 5^2}$
- ● **B.** $\sqrt{4^2 + 9^2}$
- ○ C. $\sqrt{4^2 + 1^2}$
- ○ D. $4 + 9$

B

Across 4, down 9: √97.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 6

Real-World Coordinate-Plane Problems; Perimeter

Independent Set 6.06 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. Find the perimeter of the triangle $(-8, -3)$, $(2, -3)$, $(-3, 9)$. It has no right angle.

36 units

Base 10; the height 12 lands at (−3, −3) and splits it 5 and 5, so each slanted side is 13.

2. Find the perimeter of the triangle $(-4, 1)$, $(2, 1)$, $(2, -4)$, to the nearest tenth.

18.8 units

6 + 5 + √61 ≈ 18.81.

3. Find the perimeter of the parallelogram $(-4, -5)$, $(2, -5)$, $(7, 7)$, $(1, 7)$.

38 units

Two sides of 6; two slanted sides, across 5 and up 12, of 13.

4. On a map in blocks, Kai's home is at $(-6, -5)$ and his school at $(3, 7)$. How much shorter is the straight line than the streets?

6 blocks

Streets 9 + 12 = 21; straight 15.

5. On a map in miles, a dock is at $(-4, 3)$ and an island at $(3, -5)$. How far apart are they, to the nearest tenth?

√113 ≈ 10.6 mi

Across 7, down 8; 49 + 64 = 113.

6. A rectangle has corners $(-3, -2)$, $(2, -2)$, $(2, 10)$ and $(-3, 10)$. How long is a diagonal?

13 units

Sides 2 − (−3) = 5 and 10 − (−2) = 12; 25 + 144 = 169.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 6

Real-World Coordinate-Plane Problems; Perimeter

Question Bank 6.06 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the perimeter of the triangle with the given corners. Round to the nearest tenth when it is not a whole number.

1. $(-1, -2)$, $(8, -2)$, $(8, 10)$

36 units

9, 12 and 15.

2. $(-3, 6)$, $(2, 6)$, $(-3, -6)$

30 units

5, 12 and 13.

3. $(-2, -3)$, $(4, -3)$, $(4, 4)$

22.2 units

6 + 7 + √85 ≈ 22.22.

4. $(-6, -3)$, $(6, -3)$, $(0, 5)$ — a triangle with no right angle

32 units

Base 12; the height 8 splits it 6 and 6, so each slanted side is 10.

For questions 5 through 7, each grid unit is one block or one meter.

5. Home is at $(-6, 2)$ and school at $(3, -10)$. How much shorter is the straight line than the streets?

6 blocks

Streets 21; straight 15.

6. On a park map in meters, a bench is at $(-2, -4)$ and a slide at $(6, -1)$. How far apart are they, to the nearest tenth?

√73 ≈ 8.5 m

Across 8, up 3.

7. How long is a diagonal of the rectangle with corners $(-3, -1)$, $(9, -1)$, $(9, 4)$ and $(-3, 4)$?

13 units

Sides 12 and 5.

Choose one.

8. What is the perimeter of the triangle with corners $(-3, -1)$, $(0, -1)$ and $(0, 6)$, to the nearest tenth?

- ○ A. 10
- ● **B. 17.6**
- ○ C. 20
- ○ D. 68

B

3 + 7 + √58 ≈ 17.62.

9. How much shorter is the straight path from $(-4, -2)$ to $(5, 10)$ than the path along the grid lines?

- ○ A. 15
- ○ B. 21
- ● **C. 6**
- ○ D. 36

C

21 along the grid lines, 15 straight.

GRADE 7 ACCELERATED · UNIT 6 · LESSON 6

Real-World Coordinate-Plane Problems; Perimeter

Question Bank - Additional 6.06 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the perimeter of the triangle with the given corners. Round to the nearest tenth when it is not a whole number.

1. $(-4, -1)$, $(1, -1)$, $(1, 11)$

30 units

5, 12 and 13.

2. $(6, -3)$, $(-3, -3)$, $(6, 9)$

36 units

9, 12 and 15.

3. $(-5, 2)$, $(3, 2)$, $(3, -1)$

19.5 units

8 + 3 + √73 ≈ 19.54.

4. $(-9, -4)$, $(9, -4)$, $(0, 8)$ — a triangle with no right angle

48 units

Base 18; the height 12 splits it 9 and 9, so each slanted side is 15.

For questions 5 through 7, each grid unit is one block or one meter.

5. Home is at $(-5, 6)$ and the store at $(7, 1)$. How much shorter is the straight line than the streets?

4 blocks

Streets 17; straight 13.

6. On a park map in meters, a bench is at $(-6, 1)$ and a fountain at $(2, -4)$. How far apart are they, to the nearest tenth?

√89 ≈ 9.4 m

Across 8, down 5.

7. How long is a diagonal of the rectangle with corners $(-4, -3)$, $(5, -3)$, $(5, 9)$ and $(-4, 9)$?

15 units

Sides 9 and 12.

Choose one.

8. What is the perimeter of the triangle with corners $(-1, 2)$, $(4, 2)$ and $(4, -5)$, to the nearest tenth?

- ○ A. 12
- ○ B. 24
- ● **C. 20.6**
- ○ D. 86

C

5 + 7 + √74 ≈ 20.60.

9. How much shorter is the straight path from $(-3, -5)$ to $(9, 0)$ than the path along the grid lines?

- ● **A. 4**
- ○ B. 17
- ○ C. 13
- ○ D. 30

A

17 along the grid lines, 13 straight.

GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

Similar Triangles; Indirect Measurement

Independent Set 6.07 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. Are triangles with sides 4, 10, 12 and 6, 15, 20 similar? Show the three ratios.

No — 6/4 = 15/10 = 1.5, but 20/12 ≈ 1.67

Two ratios agree and the third does not.

2. Triangle PQR is similar to triangle XYZ. PQ is 8, QR is 10 and PR is 13, and XY is 20. Find YZ and XZ.

YZ = 25 and XZ = 32.5

The names match PQ with XY: k = 20 ÷ 8 = 2.5; YZ matches QR and XZ matches PR.

3. A triangle has sides 15, 25 and 30. A similar triangle's longest side is 6. Find its other two sides.

3 and 5

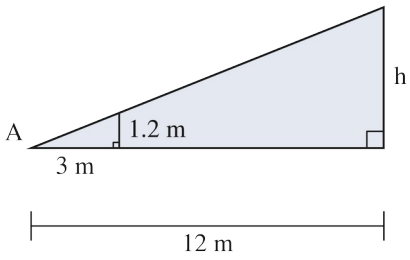
k = 6 ÷ 30 = 0.2.

4. A 1.5-m child casts a 2-m shadow. At the same moment a tree casts a 14-m shadow. How tall is the tree?

10.5 m

k = 14 ÷ 2 = 7; 1.5 × 7.

5. The two right triangles are similar and share corner A. The small one runs 3 m along the ground from A and stands 1.2 m high. The big one runs 12 m along the ground from A. How high is it?



4.8 m

Both from A: k = 12 ÷ 3 = 4, and 1.2 × 4 = 4.8.

6. Which is similar to a triangle with sides 8, 10 and 14: one with sides 10, 12, 16, or one with sides 12, 15, 21? Show why.

12, 15, 21 — every side × 1.5

10, 12, 16 adds 2 to every side.

GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

Similar Triangles; Indirect Measurement

Question Bank 6.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 and 2, decide whether the triangles are similar. Show the three ratios.

1. Sides 5, 7, 8 and 15, 21, 24

Similar — every ratio is 3

15/5 = 21/7 = 24/8 = 3.

2. Sides 6, 9, 10 and 8, 12, 15

Not similar — 8/6 = 12/9, but 15/10 = 1.5

Two ratios are 4/3; the third is 3/2.

For questions 3 through 5, the two triangles are similar. Find the missing sides.

3. Sides 4, 9 and 11; the copy's 4-side is 10.

22.5 and 27.5

k = 2.5.

4. Sides 18, 24 and 30; the copy's 30-side is 5.

3 and 4

k = 5/30 = 1/6.

5. Sides 3, 7 and 8; the copy's 7-side is 28.

12 and 32

k = 4.

For questions 6 and 7, the triangles are similar.

6. A 6-ft man's shadow is 8 ft long. At the same moment a pole's shadow is 20 ft. How tall is the pole?

15 ft

k = 20/8 = 2.5; 6 × 2.5.

7. Ty's eyes are 4 ft up, 2 ft from a mirror on the ground. In it he sees the top of a tree 15 ft beyond the mirror. How tall is the tree?

30 ft

k = 15/2 = 7.5; 4 × 7.5.

Choose one.

8. Which triangle is similar to one with sides 4, 6 and 9?

- ○ A. 6, 8, 11
- ● **B. 8, 12, 18**
- ○ C. 8, 12, 16
- ○ D. 4, 6, 12

B

Every side × 2.

9. A 5-ft girl casts a 4-ft shadow. At the same moment a tree's shadow is 24 ft. How tall is the tree?

- ○ A. 19.2 ft
- ● **B. 30 ft**
- ○ C. 25 ft
- ○ D. 120 ft

B

$k = 24 \div 4 = 6$; 5×6.

GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

Similar Triangles; Indirect Measurement

Question Bank - Additional 6.07 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 and 2, decide whether the triangles are similar. Show the three ratios.

1. Sides 6, 8, 9 and 15, 20, 22.5

Similar — every ratio is 2.5

15/6 = 20/8 = 22.5/9 = 2.5.

2. Sides 4, 5, 7 and 6, 8, 10.5

Not similar — 6/4 = 1.5, but 8/5 = 1.6

The middle ratio differs.

For questions 3 through 5, the two triangles are similar. Find the missing sides.

3. Sides 5, 6 and 8; the copy's 5-side is 15.

18 and 24

k = 3.

4. Sides 20, 30 and 35; the copy's 20-side is 4.

6 and 7

k = 0.2.

5. Sides 2, 5 and 6; the copy's 6-side is 21.

7 and 17.5

k = 3.5.

For questions 6 and 7, the triangles are similar.

6. A 1.8-m woman's shadow is 1.2 m long. At the same moment a building's shadow is 16 m. How tall is the building?

24 m

Each height is 1.8 ÷ 1.2 = 1.5 times its shadow: 16 × 1.5.

7. Mia's eyes are 5 ft up, 2 ft from a mirror on the ground. In it she sees the top of a building 30 ft beyond the mirror. How tall is the building?

75 ft

k = 15; 5 × 15.

Choose one.

8. Which triangle is similar to one with sides 3, 5 and 6?

- ● **A. 9, 15, 18**
- ○ B. 5, 7, 8
- ○ C. 6, 10, 14
- ○ D. 3, 5, 9

A

Every side × 3.

9. A 6-ft man casts a 5-ft shadow. At the same moment a tree's shadow is 35 ft. How tall is the tree?

- ○ A. 29.2 ft
- ○ B. 36 ft
- ○ C. 210 ft
- ● **D. 42 ft**

D

$k = 35 \div 5 = 7$; 6×7.

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 1

Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

Independent Set 6.T1 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. What is the complement of 61°?

29°

90 − 61.

2. What is the supplement of 76°?

104°

180 − 76.

3. Two lines cross, and one of the four angles is 58°. Find the other three.

122°, 58° and 122°

The vertical angle is equal; each neighbour makes 180 with it.

4. A linear pair measures $(5x + 10)^\circ$ and 70°. Find x and the first angle.

x = 20; 110°

5x + 80 = 180.

5. Vertical angles measure $(7x - 9)^\circ$ and $(3x + 31)^\circ$. Find the measure of each angle.

61°

7x − 9 = 3x + 31, x = 10.

6. $\angle PQR$ and $\angle RQS$ share the side QR and together make a straight line, and $m\angle PQR = 65^\circ$. Name the pair and find $m\angle RQS$.

A linear pair (adjacent and supplementary); 115°

180 − 65.

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 1

Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

Question Bank 6.T1 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the angle.

1. The complement of 17°

73°

90 − 17.

2. The supplement of 95°

85°

180 − 95.

3. The angle vertical to a 128° angle

128°

Vertical angles are equal.

4. The angle that makes a linear pair with a 41° angle

139°

180 − 41.

For questions 5 through 7, find x and the angles.

5. $(x+12)^\circ$ and $(2x)^\circ$ are complementary.

x = 26; 38° and 52°

3x + 12 = 90.

6. $(4x-6)^\circ$ and $(2x+18)^\circ$ are a linear pair.

x = 28; 106° and 74°

6x + 12 = 180.

7. Vertical angles measure $(8x-7)^\circ$ and $(5x+20)^\circ$.

x = 9; each 65°

3x = 27.

Choose one.

8. Angles M and N are supplementary, and angle M is 64°. What is angle N?

- ○ A. 26°
- ● **B. 116°**
- ○ C. 64°
- ○ D. 244°

B

180 − 64.

9. Vertical angles measure $(3x+8)^\circ$ and $(5x-14)^\circ$. A student answers 11. What did the student find?

- ○ A. the angle
- ● **B. the value of x**
- ○ C. the supplement
- ○ D. the complement

B

x = 11; each angle is 3(11) + 8 = 41°.

Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

Question Bank - Additional 6.T1 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the angle.

1. The complement of 43°

47°

90 − 43.

2. The supplement of 151°

29°

180 − 151.

3. The angle vertical to a 66° angle

66°

Vertical angles are equal.

4. The angle that makes a linear pair with a 117° angle

63°

180 − 117.

For questions 5 through 7, find x and the angles.

5. $(3x)^\circ$ and $(x+18)^\circ$ are complementary.

x = 18; 54° and 36°

4x + 18 = 90.

6. $(5x+4)^\circ$ and $(3x-8)^\circ$ are a linear pair.

x = 23; 119° and 61°

8x − 4 = 180.

7. Vertical angles measure $(9x-15)^\circ$ and $(4x+30)^\circ$.

x = 9; each 66°

5x = 45.

Choose one.

8. Angles E and F are complementary, and angle E is 37°. What is angle F?

- ○ A. 143°
- ○ B. 37°
- ○ C. 127°
- ● **D. 53°**

D

90 − 37.

9. Vertical angles measure $(6x-5)^\circ$ and $(2x+27)^\circ$. A student answers 8. What did the student find?

- ● **A. the value of x**
- ○ B. the angle
- ○ C. the complement
- ○ D. the supplement

A

x = 8; each angle is 6(8) − 5 = 43°.

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 2

Interior and Exterior Angles of a Triangle

Independent Set 6.T2 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. Two angles of a triangle are 63° and 79°. Find the third angle.

38°

180 − 142.

2. An isosceles triangle's two base angles are equal. The angle between its two equal sides is 40°. Find each base angle.

70°

(180 − 40) ÷ 2.

3. A triangle's remote interior angles are 46° and 58°. Find the exterior angle.

104°

46 + 58.

4. An exterior angle of a triangle is 140°. One remote interior angle is 62°. Find the other.

78°

140 − 62.

5. A triangle's angles are $(2x)^\circ$, $(3x)^\circ$ and $(4x)^\circ$. Find each angle.

40°, 60° and 80°

9x = 180, x = 20.

6. An exterior angle of a triangle measures $(4x+10)^\circ$; its remote interior angles measure 50° and $(2x)^\circ$. Write an equation and find x.

4x + 10 = 50 + 2x; x = 20

2x = 40.

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 2

Interior and Exterior Angles of a Triangle

Question Bank 6.T2 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the angle.

1. The third angle of a triangle with angles of 34° and 101°

45°

180 − 135.

2. The third angle of a right triangle with an angle of 27°

63°

180 − 90 − 27.

3. The exterior angle whose remote interior angles are 52° and 61°

113°

52 + 61.

4. The other remote interior angle, when an exterior angle is 118° and one remote angle is 47°

71°

118 − 47.

For questions 5 through 7, find x and the angles.

5. A triangle's angles are x°, $(x+30)^\circ$ and $(2x-10)^\circ$.

x = 40; 40°, 70° and 70°

4x + 20 = 180.

6. An isosceles triangle's base angles are each $(3x+5)^\circ$ and the third angle is 50°.

x = 20; 65° and 65°

6x + 10 + 50 = 180.

7. An exterior angle is $(6x-4)^\circ$; its remote interior angles are $(2x+8)^\circ$ and 40°.

x = 13; the exterior angle is 74°

6x − 4 = 2x + 48.

Choose one.

8. A triangle's angles are 30°, 70° and 80°. What is the exterior angle beside the 80°?

- ○ A. 80°
- ○ B. 280°
- ● **C. 100°**
- ○ D. 70°

C

30 + 70.

9. Which equation finds x when an exterior angle is $(4x)^\circ$ and its remote interior angles are 45° and $(x+15)^\circ$?

- ○ A. $4x+x+15=180$
- ● **B.** $4x=45+x+15$
- ○ C. $4x=x+15$
- ○ D. $4x+45+x+15=180$

B

Outside equals the two far ones: x = 20.

Interior and Exterior Angles of a Triangle

Question Bank - Additional 6.T2 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the angle.

1. The third angle of a triangle with angles of 29° and 88°

63°

180 − 117.

2. The third angle of a right triangle with an angle of 44°

46°

180 − 90 − 44.

3. The exterior angle whose remote interior angles are 33° and 79°

112°

33 + 79.

4. The other remote interior angle, when an exterior angle is 125° and one remote angle is 68°

57°

125 − 68.

For questions 5 through 7, find x and the angles.

5. A triangle's angles are x°, $(2x+15)^\circ$ and $(x+25)^\circ$.

x = 35; 35°, 85° and 60°

4x + 40 = 180.

6. An isosceles triangle's base angles are each $(4x-2)^\circ$ and the third angle is 64°.

x = 15; 58° and 58°

8x − 4 + 64 = 180.

7. An exterior angle is $(5x+2)^\circ$; its remote interior angles are $(3x+10)^\circ$ and 36°.

x = 22; the exterior angle is 112°

2x = 44.

Choose one.

8. A triangle's angles are 25°, 65° and 90°. What is the exterior angle beside the 65°?

- ● **A. 115°**
- ○ B. 65°
- ○ C. 295°
- ○ D. 25°

A

25 + 90.

9. Which equation finds x when an exterior angle is $(3x+6)^\circ$ and its remote interior angles are 54° and $(x)^\circ$?

- ○ A. $3x+6=x$
- ○ B. $3x+6+x=180$
- ○ C. $3x+6+54+x=180$
- ● **D.** $3x+6=54+x$

D

Outside equals the two far ones: x = 24.

GRADE 7 ACCELERATED · UNIT 6

Unit 6 Review

Relationships in Triangles **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target of Lessons 6.01–6.08 — more than the test does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part. Thread B's two angle days come after the test and are not on it.

The Triangle Inequality and the Converse

Lessons 6.01–6.02 · MA.8.GR.1.3 · 4 questions

1. Can the three lengths be the sides of a triangle? Show the test.

a. 5 cm, 9 cm and 15 cm

No — 5 + 9 = 14 is less than 15.

the two shorter sides against the longest

b. 6 inches, 10 inches and 16 inches

No — 6 + 10 = 16 is equal to the longest side, so the sides lie flat.

equal is not more

c. 11 m, 4 m and 8 m

Yes — 4 + 8 = 12 is more than 11.

the longest is listed first; 11 + 4 tests the wrong pair

2. Two sides of a triangle are 6 ft and 14 ft. Write an inequality for every possible length x of the third side.

$8 < x < 20$

between the difference, 14 − 6, and the sum, 14 + 6 — never equal to either

3. Is a triangle with these sides a right triangle? Show the test.

a. 65, 33 and 56

Yes — $33^2 + 56^2 = 1{,}089 + 3{,}136 = 4{,}225 = 65^2$.

the longest, 65, is c, though it is listed first

b. 7, 8 and 11

No — $49 + 64 = 113$, and $11^2 = 121$.

a triangle ($7 + 8 > 11$), but not a right one

4. Select ALL of the sets of side lengths that make a right triangle.

- ☐ A. 6, 7, 8
- ☒ **B. 10, 24, 26**
- ☐ C. 3, 4, 7
- ☐ D. 8, 9, 12
- ☒ **E. 16, 30, 34**

B and E

100 + 576 = 676 = 26^2 and 256 + 900 = 1,156 = 34^2: both, and nothing else.

The Pythagorean Theorem

Lesson 6.03 · MA.8.GR.1.1 · 6 questions

5. A right triangle has legs of 12 ft and 5 ft. How long is the hypotenuse?

13 ft

144 + 25 = 169

6. A right triangle has legs of 3 inches and 13 inches. How long is the hypotenuse, exactly and to the nearest tenth?

√178 in ≈ 13.3 in

9 + 169 = 178

7. A right triangle has a hypotenuse of 85 cm and one leg of 84 cm. How long is the other leg?

13 cm

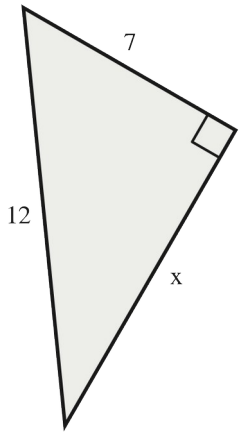
7,225 − 7,056 = 169

8. A right triangle has a hypotenuse of 11 ft and one leg of 8 ft. How long is the other leg, exactly and to the nearest tenth?

√57 ft ≈ 7.5 ft

121 − 64 = 57

9. Which equation can be used to find x?



- ○ A. $7^2 + 12^2 = x^2$
- ○ B. $7 + x = 12$
- ● **C.** $7^2 + x^2 = 12^2$
- ○ D. $7^2 + x^2 = 12$

C

The 12 is across from the right angle, so it is c and stands alone: $x^2 = 144 - 49 = 95$.

10. Ana finds the missing leg of a right triangle whose hypotenuse is 5 cm and whose other leg is 3 cm. She writes $3^2 + 5^2 = 34$, so the leg is about 5.8 cm. What did she do wrong? Find the leg.

She added the squares, and a leg cannot be longer than the 5-cm hypotenuse. Subtract: 25 − 9 = 16, so the leg is 4 cm.

the error named and the leg found

The Pythagorean Theorem in Real-World Problems

Lesson 6.04 · MA.8.GR.1.1 · 5 questions

11. A 10-ft ladder leans against a wall with its foot 6 ft from the wall. How high up the wall does it reach?

8 ft

the ladder is c: 100 − 36 = 64

12. A rectangular garden is 10 m long and 11 m wide. How long is a path straight across it from corner to corner, to the nearest tenth of a meter?

About 14.9 m (√221 m)

the diagonal is c: 100 + 121 = 221

13. A ramp rises 1 ft over a run of 9 ft along the ground. How long is the ramp, to the nearest tenth of a foot?

About 9.1 ft (√82 ft)

the ramp is c: 1 + 81 = 82

14. A park is a rectangle 9 m by 11 m. How much shorter is it to walk straight across the park from corner to corner than to walk along two of its sides? Give the answer to the nearest tenth.

About 5.8 m

along two sides 20 m; across √202 ≈ 14.2 m

15. A 9-ft ladder stands with its foot 2 ft from a wall. Will it reach a window ledge 8.5 ft up the wall?

Yes — it reaches √77 ≈ 8.8 ft up the wall, above 8.5 ft.

the height is a leg: 81 − 4 = 77

Distance on the Coordinate Plane

Lesson 6.05 · MA.8.GR.1.2 · 5 questions

16. Find the distance between the two points.

a. (−6, 2) and (5, 2)

11 units

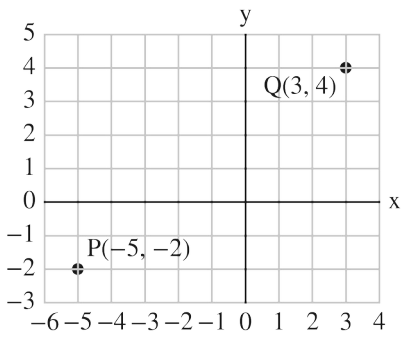
across zero: 6 + 5

b. (3, −7) and (3, 2)

9 units

across zero: 7 + 2

17. Find the distance between points P and Q.



10 units

across 8, up 6: 64 + 36 = 100

18. Find the distance between (−6, −1) and (1, 5), to the nearest tenth.

About 9.2 units (√85)

across 7, up 6: 49 + 36 = 85

19. How far is the point (−1, 6) from the origin? Round your answer to the nearest tenth.

About 6.1 units (√37)

legs 1 and 6: 1 + 36 = 37

20. What is the distance between (−2, −6) and (−2, 3)?

- ○ A. 3 units
- ● **B. 9 units**
- ○ C. 10 units
- ○ D. −9 units

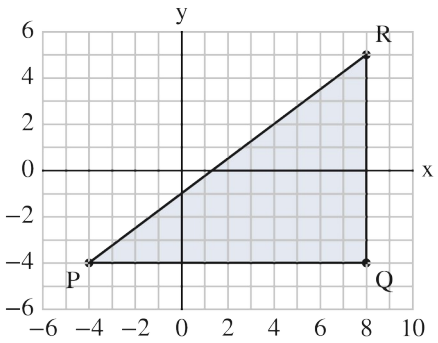
B

3 − (−6) = 9: 6 to the axis and 3 past it.

Coordinate-Plane Problems and Perimeter

Lesson 6.06 · MA.8.GR.1.2 · 4 questions

21. SSDD. Every part below is about one triangle on the grid: P(−4, −4), Q(8, −4), R(8, 5). Same surface, four different questions: a distance on a line (across zero), whether the corner at Q is square, the slanted side as a hypotenuse, and a similar copy (which needs QR, the shortest side, first) — the unit's moves on one triangle, so that a student learns to read the question and not the picture.



a. Find the length of PQ.

12 units

across zero: 4 + 8

b. Is triangle PQR a right triangle? How do you know?

Yes — PQ runs straight across and QR straight up, so they meet at a right angle at Q.

a horizontal side and a vertical side meet square (and $12^2 + 9^2 = 15^2$)

c. Find the length of PR.

15 units

the hypotenuse of legs PQ = 12 and QR = 5 − (−4) = 9: 144 + 81 = 225

d. A triangle similar to PQR has a shortest side of 3. How long is its longest side?

5

QR = 9 is the shortest side: k = 3 ÷ 9 = 1/3, and 15 × 1/3 — a copy smaller than the original

22. Find the perimeter of the triangle with corners (−4, 2), (6, 2) and (6, −1), to the nearest tenth.

About 23.4 units

10 + 3 + √109; √109 ≈ 10.440 — round once, at the end

23. On a town map in blocks, the school is at (−2, −4) and the park is at (2, 1).

a. How far is it from the school to the park along the streets?

9 blocks

4 across + 5 up

b. How far is it in a straight line? Round your answer to the nearest tenth.

About 6.4 blocks (√41)

16 + 25 = 41

24. A rectangle has corners (−3, −2), (5, −2), (5, 4) and (−3, 4).

a. Find its perimeter.

28 units

8 + 6 + 8 + 6

b. Find the length of one of its diagonals.

10 units

64 + 36 = 100

Similar Triangles and Indirect Measurement

Lessons 6.07–6.08 · MA.8.GR.2.4 · 6 questions

25. Are the two triangles similar? Show the three ratios.

a. Sides 5, 7, 8 and sides 15, 21, 24

Yes — 15/5 = 21/7 = 24/8 = 3.

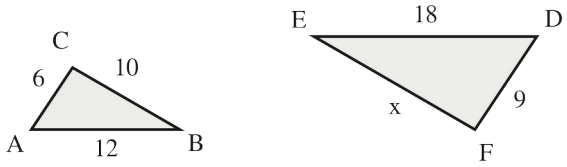
one k for every side

b. Sides 5, 8, 10 and sides 10, 16, 18

No — 10/5 = 16/8 = 2, but 18/10 = 1.8.

two equal ratios are not enough

26. Triangle ABC is similar to triangle DEF. Find x.



x = 15

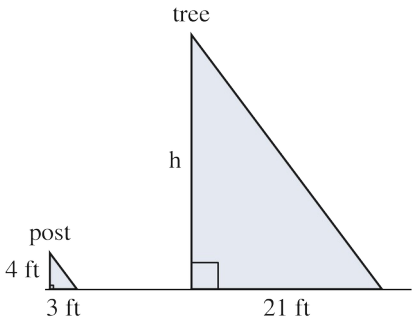
The names match AC with DF (6 with 9) and BC with EF, where x is: k = 1.5 and x = 10 × 1.5.

27. A triangle has sides 15, 20 and 30. A similar triangle's middle side is 8. Find its other two sides.

6 and 12

k = 8 ÷ 20 = 0.4 — the copy is smaller

28. A 4-ft fence post casts a 3-ft shadow. At the same moment, a tree casts a 21-ft shadow. How tall is the tree?



28 ft

k = 21 ÷ 3 = 7 from the shadows; 4 × 7

29. Kai's eyes are 5 ft above the ground. He stands 3 ft from a mirror on the ground and sees in it the top of a pole that stands 12 ft beyond the mirror. The two right triangles are similar. How tall is the pole?

20 ft

k = 12 ÷ 3 = 4; 5 × 4

30. Which triangle is similar to one with sides 6, 8 and 9?

- ○ A. 8, 10, 11
- ○ B. 12, 16, 20
- ● **C. 9, 12, 13.5**
- ○ D. 6, 8, 12

C

Every side × 1.5.

GRADE 7 ACCELERATED · UNIT 6 · PRACTICE TEST

Relationships in Triangles

18 points **ANSWER KEY**

Name
Date
Period

Show your work. One point for each lettered part.

Triangle Inequality and the Converse

3 points MA.8.GR.1.3

1. Which set of side lengths can form a triangle? *1 pt*

- ○ A. 4 cm, 7 cm, 11 cm
- ○ B. 12 cm, 5 cm, 6 cm
- ● **C. 6 cm, 8 cm, 13 cm**
- ○ D. 3 cm, 10 cm, 15 cm

C

The two shorter sides against the longest: 6 + 8 = 14, more than 13. A lies flat; B passes only when the first two listed are added; D's two shorter sides cannot reach across 15.

Scoring: 1 point.

2. Select ALL of the sets of side lengths that make a right triangle. *1 pt*

- ☒ **A. 17, 8, 15**
- ☐ B. 5, 6, 7
- ☒ **C. 20, 21, 29**
- ☐ D. 7, 24, 26
- ☐ E. 4, 6, 8

A and C

Square all three and test the longest as c: 64 + 225 = 289 = 17^2 (the longest listed first) and 400 + 441 = 841 = 29^2. Both and nothing else for the point.

Scoring: 1 point.

3. Two sides of a triangle are 6 inches and 15 inches long. Which inequality shows every possible length x, in inches, of the third side? *1 pt*

- ○ A. $6 < x < 15$
- ● **B.** $9 < x < 21$
- ○ C. $9 \le x \le 21$
- ○ D. $0 < x < 21$

B

15 − 6 = 9 and 15 + 6 = 21, never equal to either. A stays between the two given sides; C counts the flat cases, 9 and 21; D has only the sum.

Scoring: 1 point.

The Pythagorean Theorem

6 points MA.8.GR.1.1

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

4. Find the missing side of each right triangle.

a. Its legs are 9 inches and 12 inches. How long is the hypotenuse? *1 pt*

15 in

$c^2 = 81 + 144 = 225$

b. Its hypotenuse is 61 meters and one leg is 60 meters. How long is the other leg? *1 pt*

11 m

a leg: subtract — 3,721 − 3,600 = 121. Adding the squares gives a 'leg' longer than the hypotenuse.

Scoring: a 1 · b 1 = 2 points.

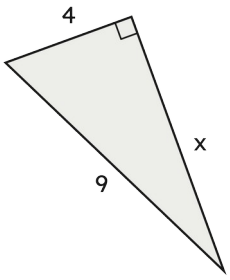
5. A right triangle has legs of 7 centimeters and 13 centimeters. How long is the hypotenuse, in centimeters? Round your answer to the nearest tenth. *1 pt*

14.8 cm (exactly √218 cm)

$c^2 = 49 + 169 = 218$; √218 = 14.76…

Scoring: 1 point.

6. Which equation can be used to find x? *1 pt*



- ○ A. $4 + x = 9$
- ○ B. $4^2 + 9^2 = x^2$
- ○ C. $4^2 + x^2 = 9$
- ● **D.** $4^2 + x^2 = 9^2$

D

The 9 is across from the right angle, so it is c and stands alone: $x^2 = 81 − 16 = 65$. A adds the sides, not their squares; B puts the hypotenuse with the legs; C leaves the 9 unsquared.

Scoring: 1 point.

7. A rectangular garden measures 9 meters by 10 meters. A path runs straight across it from one corner to the opposite corner.

a. How long is the path, in meters? Round your answer to the nearest tenth. *1 pt*

13.5 m (exactly √181 m)

the path is the hypotenuse: 81 + 100 = 181

b. How much shorter is the path than walking along two sides of the garden? Round your answer to the nearest tenth. *1 pt*

5.5 m

9 + 10 = 19 m along two sides; 19 − 13.5. Follow-through on part (a).

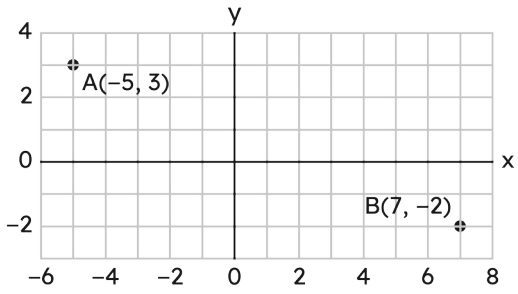
Scoring: a 1 · b 1 = 2 points.

Distance on the Coordinate Plane

6 points MA.8.GR.1.2

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

8. Find the distance between points A and B. *1 pt*



13 units

Across 7 − (−5) = 12, down 3 − (−2) = 5: c^2 = 144 + 25 = 169.

Scoring: 1 point.

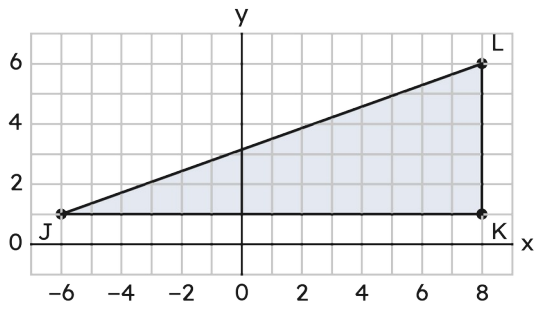
9. Find the distance between (−1, 3) and (4, 1). Round your answer to the nearest tenth. *1 pt*

5.4 units (exactly √29)

Across 5, down 2: c^2 = 25 + 4 = 29.

Scoring: 1 point.

10. Triangle JKL has corners J(−6, 1), K(8, 1) and L(8, 6).



a. Find the length of side JK. *1 pt*

14 units

Across zero: 6 + 8 — never 8 − 6.

b. Find the perimeter of triangle JKL. Round your answer to the nearest tenth. *1 pt*

33.9 units

KL = 5; JL = √(196 + 25) = √221 ≈ 14.866; 14 + 5 + 14.866 = 33.866 — round once, at the end. Follow-through on part (a).

Scoring: a 1 · b 1 = 2 points.

11. On a town map, each unit is one block. The library is at (−1, −5) and the pool is at (8, 7).

a. How far is it from the library to the pool along the streets? *1 pt*

21 blocks

The streets follow the grid: 9 across and 12 up.

b. How far is it in a straight line? *1 pt*

15 blocks

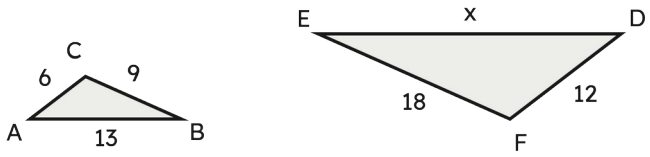
The hypotenuse of legs 9 and 12: 81 + 144 = 225.

Scoring: a 1 · b 1 = 2 points.

Similar Triangles

3 points MA.8.GR.2.4

12. Triangle ABC is similar to triangle DEF. Find x. *1 pt*

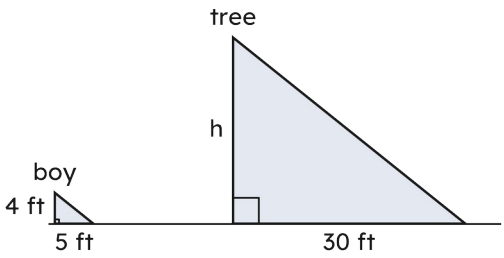


x = 26

The names match AC with DF and BC with EF — 12 with 6 and 18 with 9, as by size — so k = 2, and x is on DE, which matches AB: 13 × 2. 39 matched by place on the page (18 with the 6); 13 took similar as equal; 19 added the difference 12 − 6.

Scoring: 1 point.

13. A boy 4 ft tall casts a shadow 5 ft long. At the same moment, a tree casts a shadow 30 ft long. How tall is the tree? *1 pt*



24 ft

k = 30 ÷ 5 = 6 from the shadows; 4 × 6. 37.5 matched height and shadow crosswise; 120 multiplied by the boy's height, not by k.

Scoring: 1 point.

14. Which triangle is similar to a triangle with sides 6, 7 and 10? *1 pt*

- ● **A. 12, 14, 20**
- ○ B. 10, 11, 14
- ○ C. 12, 14, 18
- ○ D. 6, 7, 12

A

Every side × 2: 12/6 = 14/7 = 20/10. B adds 4 to every side; C passes two ratios but not the third, 18 ÷ 10; D keeps two sides the same.

Scoring: 1 point.

Score Tracker

Benchmark	Questions	Possible	Earned
MA.8.GR.1.3	1, 2, 3	3	
MA.8.GR.1.1	4, 5, 6, 7	6	
MA.8.GR.1.2	8, 9, 10, 11	6	
MA.8.GR.2.4	12, 13, 14	3	
Total		**18**	

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page. Nothing is weighed and nothing is arguable: apply the part's operation to the student's own earlier value and tick if that is what is written.